

SIMATS ENGINEERING

Department of Computer Science and Engineering

Course Code:	CSA15112	Assessment:	CO5 AT2
Course Name:	Cloud Computing and Big Data Analytics	Type:	Operations Optimization
CO Statement:	Apply big data storage, integration, Hadoop, HDFS, and MapReduce concepts for distributed data processing		

Part A: Production Hadoop Operations – Lesson Notes

Monitoring & Observability

A production Hadoop cluster generates thousands of metrics. Monitoring means: (1) collecting these metrics, (2) storing them, (3) displaying dashboards, (4) setting thresholds to trigger alerts.

Component	Key Metrics	Normal Range	Alert Threshold
NameNode	Heap usage, Live/Dead DataNodes, Block count, Edit log size	< 70% heap	> 85% heap
DataNode	Disk usage, Block count, Network I/O, JVM heap	< 80% disk	> 92% disk
MapReduce	Job completion rate, Task failure %, Shuffle sort time	< 5% failure	> 20% failure
HDFS	Under-replicated blocks, Missing blocks, Block replica imbalance	0 under-replicated	> 10 blocks

Lesson Note: Why Monitoring Matters

Without monitoring, you only discover problems when users complain. NameNode heap at 95% → crash → cluster down → data unavailable. Monitoring alerts at 85% gives you time to act (add DataNodes, scale up, optimize)

Backup & Disaster Recovery

RTO (Recovery Time Objective): Maximum acceptable downtime during an outage.

RPO (Recovery Point Objective): Maximum acceptable age of data loss upon recovery.

Key Concepts:

1. RTO (Recovery Time Objective): How long can the cluster be down? 1 hour? 1 minute?
2. RPO (Recovery Point Objective): How much data loss is acceptable? 1 hour? 1 day?

Backup Strategy	RTO	RPO	Cost	Best For
No backup (risky)	∞ (unrecoverable)	∞	\$0	Dev/test only
Daily full backup to external disk	1–2 hours	24 hours	\$100/month	Dev, non-critical
Hourly incremental + offsite replication	15 minutes	1 hour	\$500/month	Production
HA (active/standby NameNode + ZK quorum)	< 1 minute	0 (near-instant)	\$1000+/month	Critical production

High Availability (HA) Design

HA eliminates the single NameNode as a single point of failure. Setup: Active NameNode + Standby NameNode + ZooKeeper quorum (3 nodes). When Active fails, Standby takes over automatically in < 1 minute. Requires shared storage (NFS/QJM) for edit logs.

Cost Optimization

Monthly cluster cost = Hardware + Power + Cooling + Ops

Cost Component	Typical \$	Control Method
Hardware (50 nodes × \$500)	\$25,000 one-time	Buy less, or decommission slow nodes
Annual power & cooling (50 nodes × \$1000)	\$50,000/year	Efficient hardware, underutilization
Operations & support	\$10,000/month	Automation, monitoring, runbooks
Compression (Gzip, Snappy)	\$-5,000/month savings	Reduced storage, higher CPU
Tiering (move old logs to cold storage)	\$-3,000/month savings	Archive logs > 30 days to S3

Scaling Strategies

Horizontal scaling: Add more DataNodes (linear cost, complexity)

Vertical scaling: Upgrade existing nodes (cap at 10–20TB/node practical limit)

NameNode Capacity Constraint

NameNode heap number of files. Rule of thumb: 1GB heap ≈ 1M files. If 100M files needed, you need 100GB NameNode heap — impractical. Solutions: (1) Tiering (archive old data), (2) Federation (multiple NameNodes), (3) Consolidate small files.

Part B: Main Scenario – Optimize a 50-Node Production Cluster

Current State Assessment

A company runs a 50-node HDFS cluster processing 10TB/day. Current metrics:

Metric	Current Value	Status
NameNode heap usage	85%	RISKY – near crash threshold
DataNode disk utilization	92%	RISKY – no space for replication
Monthly cost	\$50,000	High, no optimization
Last backup	2 weeks ago	UNACCEPTABLE RPO
DataNode failure recovery time	8 hours	UNACCEPTABLE RTO
Monitoring setup	Manual spreadsheets	No real-time alerts
HA setup	None (single NameNode)	Single point of failure

Problem Summary

The cluster is on the edge of failure. NameNode heap crash is imminent. No automated failover. Slow recovery from node failures. Costs unchecked. Backup is stale.

Step 1: Design Monitoring Dashboard

Create a monitoring strategy: what to track, alert thresholds, and dashboard design.

1a. NameNode Metrics

Metrics to track:

- Heap usage (%)
- Live/Dead DataNodes (count)
- Block count (total)
- Edit log size (MB)

Metric	Green	Yellow	Red
NameNode heap %	< 60%	60–80%	> 80%
Dead DataNodes	0	1–2	> 2
Under-replicated blocks	0	< 100	> 100

Fill in your thresholds and actions:

Trigger alert at 80%; auto-archive log directories older than 30 days.
Rebalance replication; pause non-critical MapReduce jobs.
Throttle ingest stream; increase HDFS block replicator priority.
Set node to read-only; trigger HDFS Balancer.

Step 2: Plan Backup & Disaster Recovery

Current state: Last backup 2 weeks ago → RPO of 14 days is unacceptable.

2a. Backup Strategy

Design a backup plan:

Decision	Option A	Option B	Option C
Backup frequency	Daily	Hourly	Real-time (HA)
Storage location	On-cluster NFS	Off-site S3	Shared QJM
RTO	2–4 hours	1 hour	< 1 minute
RPO	24 hours	1 hour	Near-zero
Cost/month	\$200	\$1000	\$2000+ HA infra

Your recommendation: Choose the option that balances cost and business needs.

- Selected Strategy: Option C (Real-Time HA via QJM + Daily Off-site S3 Snapshots)
- Target RTO / RPO: RTO < 1 minute (via automatic ZK failover); RPO near-zero for metadata.
- Justification: Avoids catastrophic downtime. The cluster processes 10 TB/day; an unrecoverable crash or multi-hour outage creates higher business losses than the \$1,000–\$2,000/month infrastructure cost.

Step 3: Cost Optimization Analysis

Current: 50 nodes × \$1000/year ops = \$50K/year. Target: Reduce by 20% to \$40K/year.

Optimization	Annual Savings	Impact	Your Choice?
Compression (Gzip logs)	\$5,000	Reduced storage, higher CPU	Yes
Tiering (archive > 30 days to cold)	\$3,000	Slower access to old logs	Yes
Decommission 5 slow nodes	\$5,000	Less capacity, need to rebalance	Yes
Auto-scaling (scale down off-peak)	\$2,000	Reduced peak hours ops	Yes

Calculate total savings from your selections and new monthly cost:

- **Total Annual Savings:** \$5,000 + \$3,000 + \$5,000 + \$2,000 = \$15,000/year (30% reduction, exceeding the 20% / \$10,000 target).
- **New Cluster Operations Cost:** \$50,000 – \$15,000 = \$35,000/year (≈ \$2,916.67/month).

Step 4: Address NameNode Heap Pressure

Current heap: 85% at 50M files. At current growth (1M files/month), heap will exceed 90% in ~2 months → crash risk.

Option	Action	NameNode Heap Impact	Timeline
Tiering	Archive logs > 30 days (removes old files)	Reduces heap by ~10%	1 month
Consolidate	Merge small files into larger blocks	Reduces heap by ~15%	Ongoing
Scale	Add second NameNode (Federation)	Splits namespace across 2 NameNodes	2 months setup

Which solution do you recommend? Why?

- Recommended Strategy: Immediate Tiering combined with continuous File Consolidation.
- *Immediate (Month 1)*: Archive logs older than 30 days to cold storage, reducing heap by $\approx 10\%$.
- *Short-Term (Ongoing)*: Deploy MapReduce archive jobs to merge small files into sequence files, freeing an additional $\approx 15\%$ heap space.
- *Long-Term (Month 3+)*: Prepare HDFS Federation if total file namespace continues to approach the 100M ceiling.

Step 5: Implement High Availability (HA)

Current: Single NameNode (SPOF). Proposed: HA with Standby.

Component	Quantity	Purpose
Active NameNode	1	Primary metadata server
Standby NameNode	1	Automatic failover replica
ZooKeeper quorum	3 or 5	Coordinates HA election
Shared edit log storage	NFS or QJM	Both NNs access same metadata

Failover process: Active NameNode fails → ZK detects (30 sec) → Standby promotes (30 sec) → Clients reconnect.
Total RTO < 1 minute.

Estimate HA setup cost and timeline:

- Failover SLA: Detection (30s) + State Promotion (30s) = Total RTO < 1 minute.
- Estimated HA Infrastructure Cost: $\approx \$1,000 - \$1,500/\text{month}$; Deployment timeline: 3 weeks.

Part C: Mini-Problems

Mini-Problem 1: DataNode Failure Scenario

A DataNode holding 12TB of data fails (hardware crash). HDFS detects the failure and begins re-replicating blocks from surviving replicas. Current network capacity: 1 Gbps per DataNode.

Calculate re-replication time and estimate when cluster is back to normal replication factor.

Given: 12 TB lost data; 1 Gbps network bandwidth per node (= 125 MB/s = 0.125 GB/s).

Conversion: 12 TB = 12,000 GB = 12,288,000 MB.

Theoretical Minimum Time (Single Node Pipe):

$$\text{Time} = \frac{12,288,000 \text{ MB}}{125 \text{ MB/s}} = 98,304 \text{ seconds} \approx 27.3 \text{ hours}$$

Mini-Problem 2: NameNode Heap Projection

Current state: 50M files, 60% heap usage. Growth rate: 1.5M files/month. If file count grows linearly, when will NameNode heap exceed 90%? When should you add a second NameNode (Federation)?

Given: 50M files corresponds to 60% heap; linear growth of 1.5M files/month.

Total Heap Capacity: $\frac{50\text{M files}}{0.60} \approx 83.33\text{M files capacity}$.

90% Heap Limit: $83.33\text{M} \times 0.90 = 75\text{M files}$.

Namespace Headroom: $75\text{M} - 50\text{M} = 25\text{M files}$.

$$\text{Time} = \frac{25\text{M files}}{1.5\text{M files/month}} = 16.67 \text{ months}$$

Mini-Problem 3: Compression Trade-off

Implement Gzip compression on 20TB of historical logs. Storage reduction: 3:1 (20TB → 6.67TB). But: CPU overhead for decompression during MapReduce jobs increases job time by 15%.

Is compression worth it economically? Calculate storage cost savings vs. increased compute cost.

Given: 20 TB storage compressed to 6.67 TB (saving 13.33 TB); 15% increase in compute execution

Monthly Storage Savings = $13.33 \text{ TB} \times \$50/\text{TB} = \$666.50/\text{month}$

Additional Compute Cost = $1,000 \text{ hrs} \times 15\% \times \$2.00/\text{hr} = \$300.00/\text{month}$

Net Benefit: $\$666.50 - \$300.00 = +\$366.50/\text{month}$ net savings. Compression is economically viable.

Mini-Problem 4: Tiering Strategy

Move logs older than 30 days to cold storage (AWS S3 Glacier, \$1/TB/month). Current hot storage cost: \$50/TB/month. Hot tier capacity: 20TB. Cold tier capacity: unlimited.

Calculate break-even point: after how many days of cold storage does it become cheaper than hot?

Given: Hot storage = \$50/TB/month (\$1.667/TB/day); Cold storage = \$1/TB/month (\$0.0333/TB/day);

S3 Glacier Lifecycle transition = \$0.05/TB one-time.

Daily Savings per TB: $\$1.667 - \$0.0333 = \$1.6337/\text{TB/day}$.

$$\text{Break-Even Days} = \frac{\text{Transition Cost}}{\text{Daily Net Savings}} = \frac{\$0.05}{\$1.6337} \approx 0.03 \text{ days (less than 1 day)}$$

Part D: Problem Bank – 4 Additional Scenarios

Scenario A: Multi-Datacenter Failover

Company wants to replicate the 50-node cluster to a second datacenter 100km away for disaster recovery. Design cross-datacenter replication strategy, network bandwidth requirements, and cost implications. Should data be synchronously or asynchronously replicated? What is RTO/RPO?

Replication Type: Asynchronous cross-datacenter replication using Apache Falcon or DistCp.
Synchronous replication over 100 km adds network latency that stalls HDFS pipeline writes.

$$\text{Bandwidth} = \frac{10 \times 10^{12} \times 8 \text{ bits}}{86,400 \text{ seconds}} \approx 926 \text{ Mbps} \quad (1 \text{ Gbps dedicated link recommended})$$

Scenario B: Federation for Scale-Out

The cluster needs to support 200M files (not possible with single NameNode due to heap limits). Design a Federation setup: how many NameNodes? How to distribute files across namespaces? What is the performance impact? Can you still run cluster-wide MapReduce jobs?

- **Architecture:** 3 NameNodes partition the metadata:
 - NameNode 1: /user namespace (70M files)
 - NameNode 2: /data/analytics namespace (70M files)
 - NameNode 3: /archive namespace (60M files)
- **Client Routing:** ViewFS client-side mount tables abstract path resolutions.

Scenario C: Real-Time Alerting & Auto-Remediation

Design a monitoring + auto-remediation system: when DataNode disk usage exceeds 85%, automatically trigger: (1) alert to ops, (2) pause new block writes to that node, (3) trigger re-replication of critical blocks. What tools would you use? How to test safely?

Toolchain: Prometheus + Grafana + JMX Exporter + Ansible Engine.

Prometheus detects DataNode storage > 85% via JMX metric

Ansible playbook triggers HDFS admin command: `hdfs dfsadmin -setSpaceQuota` to throttle ingest on that node.

Orchestrates background hdfs balancer `-threshold 5` to redistribute blocks safely without manual intervention.

Scenario D: Cost Benchmarking & Optimization

50 nodes hardware amortization (\$25,000) + Power/Cooling (\$100,000) + 10% Ops (\$24,000)

100 TB @ \$23/TB/month × 24 months + Ingress/Egress (\$1,200)

20 TB Hot On-Prem (\$29,800) + 80 TB Cold Glacier (\$1/TB/mo = \$1,920) + 10% Ops (\$24,000)

The company wants to understand their true cost per TB stored. Include: hardware amortization, power, cooling, network, ops salary (allocate 10% of an engineer's time). Compare: (1) on-premise cluster, (2) AWS S3, (3) hybrid (hot on-premise + cold cloud). Which is most cost-effective for 100TB with 2-year horizon?

Evaluation Rubric

Criteria	Wt.	L4 (40–50)	L3 (30–39)	L2 (20–29)	L1 (0–19)
Monitoring & Observability	20%	Comprehensive metrics, realistic alert thresholds, actionable	Good metrics; some thresholds arbitrary	Basic metrics; incomplete alert strategy	Incomplete monitoring design
Cost Optimization	25%	Accurate cost breakdown, realistic savings targets, trade-offs justified	Good analysis; minor gaps in cost modeling	Basic cost analysis; limited optimization ideas	Incorrect or missing cost analysis
Backup & DR Planning	20%	Credible RTO/RPO targets, feasible failover design, HA understanding	Good DR plan; some RTO/RPO unrealistic	Basic backup strategy; weak failover plan	Incomplete or missing strategy
Scaling & Capacity Planning	20%	Data-driven forecasts, proactive scaling, Federation/HA impact analysis	Good capacity plan; some growth projections weak	Basic capacity analysis; limited forward planning	Missing or unrealistic capacity planning
Documentation & Clarity	15%	Clear dashboards, formulas, step-by-step reasoning, justified decisions	Mostly clear; minor formatting/detail gaps	Understandable but lacks design diagrams/detail	Unclear or poorly organized

Total Marks: 100 (20 + 25 + 20 + 20 + 15)